

Karthigan Mageswaran

First-Class Computer Science Graduate

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PROFILE

First-Class Computer Science graduate from the University of Leicester with practical experience in backend system architecture and cloud computing concepts. Currently studying for the AWS Certified Solutions Architect – Associate (SAA) qualification. Experienced in operational management, dynamic resource allocation, and technical support. Trilingual in English, Tamil, and Italian, bringing excellent communication and problem-solving skills ideal for enterprise IT service delivery, infrastructure management, and field engineering roles. Actively seeking a junior role in cloud infrastructure, IT service delivery, or field engineering to apply technical troubleshooting skills in an enterprise IT environment.

EDUCATION

BSc Computer Science · University of Leicester

Sep 2023 - Jul 2026

First-Class Honours

Relevant modules: Internet and Cloud Computing, Operating Systems & Networks, Foundations of Cyber Security, Databases & Domain Modelling, Software Architecture & System Development, Data Analytics.

TECHNICAL SKILLS

- **Infrastructure & Cloud :** AWS Certified Solutions Architect - Associate (SAA) (In Progress), Server-side architecture, asynchronous event pipelines, API integrations, Internet and Cloud Computing concepts.
- **Networking & Security:** JWT, BCrypt, PCI-DSS compliance frameworks.
- **Database & Data Management:** MySQL, Spring Data JPA / Hibernate, Relational Database Design.
- **Systems & Automation:** Bash / Shell, SQL, Git / GitHub, Postman, Agile
- **Programming & Scripting:** Python, Java, HTML5
- **Languages:** English (fluent), Tamil (native), Italian (native).

PROJECTS

University Accommodation Allocation System · Final Year Project

2024 – 2026

Spring Boot 3.2 · React 18 · MySQL · Spring Security · Stripe · OpenAI (Spring AI) · Google Maps · Tailwind CSS · OpenPDF

- Designed and implemented a four-stage hybrid allocation engine (SITS validation, group clustering, greedy preference matching, weighted penalty matrix) processing 25 concurrent applications in under 5 seconds.
- Implemented role-based access control across three user types (Student, Site Manager, Admin) with server-side strict role-based access control, BCrypt credential hashing, and JWT-based session management.
- Integrated Stripe Elements for secure payment processing achieving PCI-DSS compliance.
- Built a GPT-4o-mini conversational assistant using Spring AI function-calling, grounded against a curated knowledge base to prevent hallucinations and enforce institutional policy boundaries.
- Delivered asynchronous event pipelines for SMTP email dispatch, server-side PDF tenancy agreement generation, and automated 72-hour offer eviction sweeps using Spring @Async and @Scheduled.

Personal Cybersecurity Dashboard · Python · Streamlit · SQLite · zxcvbn-python

2024

- Built a standalone security monitoring tool assessing password entropy using the zxcvbn algorithm and simulating email breach threat-intelligence tracking.
- Implemented a local SQLite database for persistent alert logging and security event validation, demonstrating applied information security principles.

IBM SkillsBuild Gamified Learning App · Java · Spring Boot · JavaScript · REST APIs · Agile team project

2023

- Collaborated in an Agile team to design and integrate an external gamification platform with IBM SkillsBuild, developing the backend for user achievement and engagement tracking.
- Responsible for ensuring secure transmission and storage of user metrics, applying RESTful API design principles throughout.

EXPERIENCE

Shift Leader · Tesco

Jan 2022 – Present

- Lead and coordinate a team across high-volume shifts, managing effective task delegation, staff welfare, and real-time escalation handling.
- Hold full keyholder responsibilities including opening/closing procedures, cash reconciliation, and physical security management
- Consistently manage competing priorities across customer service, stock operations, and delivery coordination in a time-pressured environment.

Freelance Interpreter - Tamil & Italian · AaGlobal Language Services

Jun 2024 – Jan 2026

- Delivered professional interpretation in high-stakes medical and legal settings, operating under strict confidentiality and professional conduct requirements.

IT Technician - Work Experience · Ollak Limited

Oct 2021 – Feb 2023

- Provided comprehensive 1st and 2nd line technical support across software, hardware, and network configuration for local businesses.
- Troubleshoot and resolved technical issues efficiently, achieving high client satisfaction rate.

C E R T I F I C A T I O N S & A C H I E V E M E N T S

- Winner, Network Rail Diverse IT Competition (2021) - selected as one of 50 winners nationwide for an innovative tech solution.
- Completed BSc degree while working two jobs.
- Certified First Aider